

SOC Security Analytics Dashboard using Splunk, Python, Streamlit & Gemini AI

Abstract

Security Operations Centers (SOCs) rely on Security Information and Event Management (SIEM) platforms to identify and respond to cyber threats. Manual investigation of large volumes of security logs is time-consuming and may delay incident response. This project presents a SOC Security Analytics Dashboard developed using Splunk Enterprise, Python, Streamlit, and Google Gemini AI. The solution ingests SSH authentication logs into Splunk, performs threat detection using SPL queries, enriches security events through a Python-based detection engine, maps attacks to the MITRE ATT&CK framework, and visualizes incidents through an interactive dashboard.

The dashboard provides threat metrics, severity distribution analysis, threat exploration capabilities, AI-assisted incident investigation, custom SPL querying, and threat report export functionality. The system successfully analyzed 1,000 SSH authentication events and identified multiple brute-force attacks, root account attacks, and successful SSH compromises. The project demonstrates the integration of SIEM, threat intelligence, visualization, and Generative AI to improve security monitoring and incident response workflows.

Keywords: SOC, SIEM, Splunk, Streamlit, Python, Cybersecurity, Threat Detection, MITRE ATT&CK, Generative AI.

1. Introduction

Organizations generate thousands of security events daily from servers, endpoints, firewalls, and applications. Security analysts must efficiently identify malicious activity from large volumes of log data. Traditional log review processes are often manual and time-consuming, increasing the risk of delayed threat detection.

The objective of this project is to design and implement a Security Operations Center (SOC) dashboard capable of detecting suspicious SSH activities, classifying threats, assigning risk scores, mapping adversary techniques to MITRE ATT&CK, and providing AI-generated investigation summaries. The dashboard serves as a centralized platform for security monitoring and incident analysis.

2. Problem Statement

Security teams face several challenges while investigating security incidents:

- Large volumes of log data
- Difficulty identifying malicious activity quickly
- Lack of contextual threat intelligence
- Manual report generation processes
- Limited visibility into attack patterns

To address these challenges, a SOC Security Analytics Dashboard was developed to automate threat detection, risk assessment, visualization, and investigation support.

3. Objectives

The primary objectives of the project are:

- Collect and analyze SSH authentication logs using Splunk
- Detect brute-force attacks and account compromise attempts
- Assign severity levels and risk scores to threats
- Map detected attacks to MITRE ATT&CK techniques
- Generate AI-assisted investigation reports
- Provide security dashboards for threat visualization
- Enable analysts to execute custom SPL searches
- Export threat reports for documentation purposes

4. Technology Stack

Component	Technology Used
SIEM Platform	Splunk Enterprise
Programming Language	Python 3.12
Dashboard Framework	Streamlit
Data Processing	Pandas
AI Analysis Engine	Google Gemini API
Threat Framework	MITRE ATT&CK
Report Export	CSV
Operating System	Windows 11

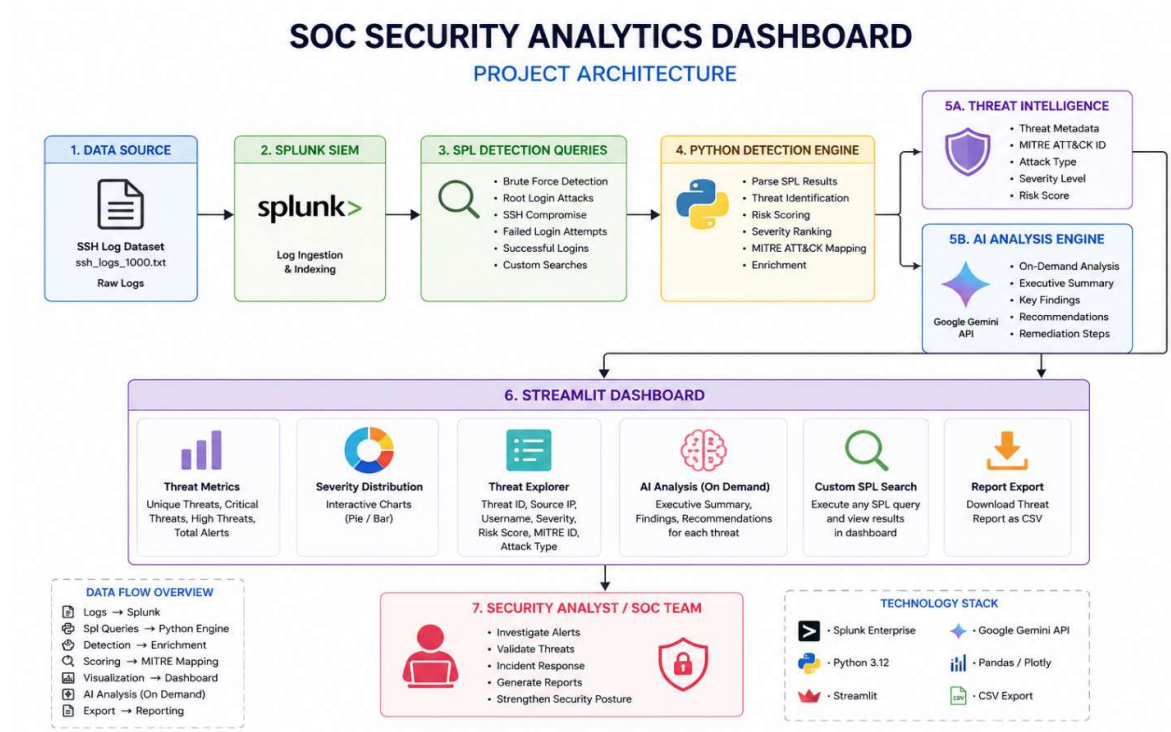
5. System Architecture

The architecture consists of six major components:

1. SSH Log Dataset
2. Splunk SIEM
3. SPL Detection Queries
4. Python Detection Engine
5. AI Analysis Engine
6. Streamlit Dashboard
- 7.

Workflow:

SSH Logs → Splunk → SPL Queries → Python Processing → Threat Intelligence Mapping
→ AI Analysis → Dashboard Visualization → CSV Reporting



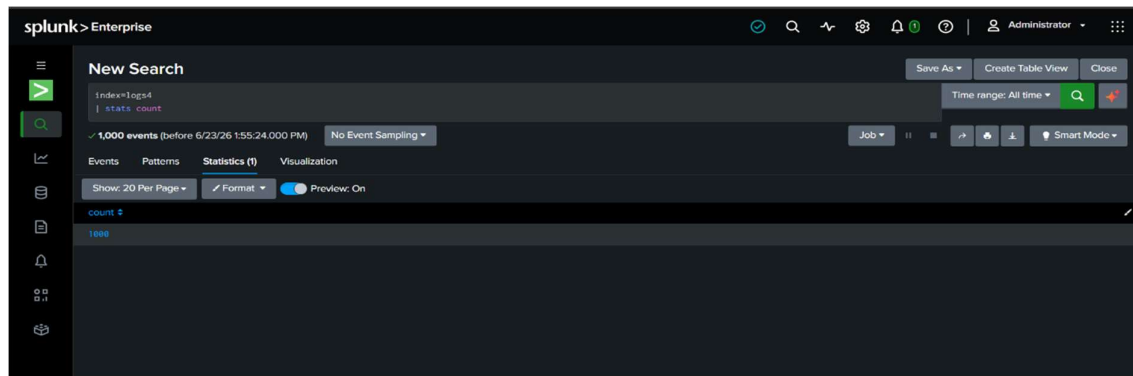
6. Dataset Description

The project utilizes a simulated SSH authentication dataset containing 1,000 log events.

Dataset Characteristics:

- Total Events: 1,000
- Log Source: SSH Authentication Logs
- Event Types:
 - Failed Login Attempts
 - SSH Brute Force Attacks
 - Root Account Attacks
 - Successful SSH Compromises
 - Fields Captured:
 - Source IP Address
 - Username
 - Event Type
 - Timestamp

The dataset was ingested into Splunk Enterprise for indexing and analysis.



7. Threat Detection Methodology

Threat detection is performed using Splunk Search Processing Language (SPL) queries.

The detection engine identifies three primary attack categories:

1)SSH Brute Force Attack

Description:

Multiple failed authentication attempts targeting a user account.

MITRE ATT&CK Mapping:

T1110 – Brute Force

Severity:

High

Risk Score:

75

2)Root Account Attack

Description:

Repeated login attempts against privileged root accounts.

MITRE ATT&CK Mapping:

T1110 – Brute Force

Severity:

Critical

Risk Score:

90

3)Successful SSH Compromise

Description:

Successful authentication following multiple failed login attempts.

MITRE ATT&CK Mapping:

T1078 – Valid Accounts

Severity:

Critical

Risk Score:

95

8. Threat Scoring Engine

The Python detection engine enriches security events by assigning severity levels and risk scores.

Severity	Risk Score
Low	25
Medium	50
High	75
Critical	95

Threat prioritization enables analysts to focus on high-risk incidents first.

9. Dashboard Features

9.1 Threat Metrics

The dashboard provides real-time security statistics:

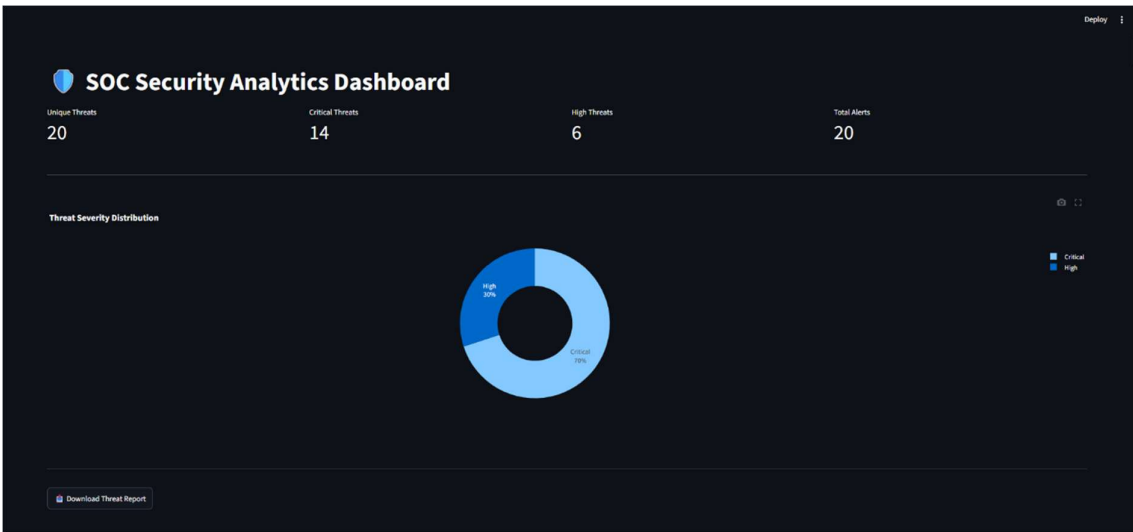
- Unique Threats: 20
- Critical Threats: 14
- High Threats: 6
- Total Alerts: 20

9.2 Severity Distribution Visualization

The dashboard visualizes the distribution of threats using interactive charts.

Purpose:

- Identify threat trends
- Understand severity distribution
- Improve security monitoring

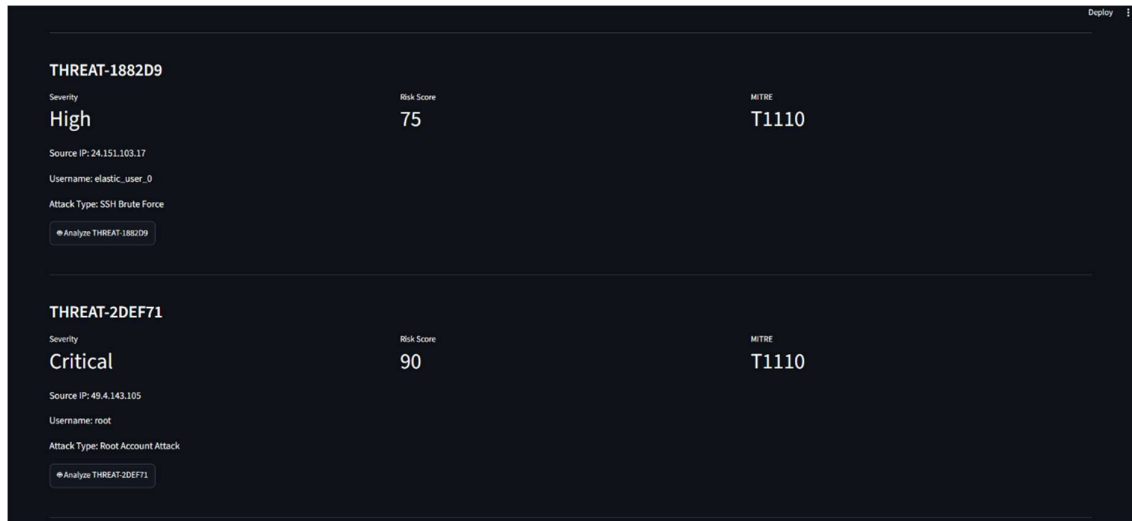


9.3 Threat Explorer

Each detected threat contains:

- Threat ID
- Source IP
- Username
- Attack Type
- Risk Score
- MITRE ATT&CK Technique

This allows analysts to investigate individual incidents in detail.



9.4 AI Security Analysis

Generative AI is integrated on-demand to assist analysts during investigations.

The AI module generates:

Executive Summary

Provides a concise overview of the incident.

Findings

Identifies suspicious behavior, affected accounts, and attack characteristics.

Recommendations

Suggests mitigation and remediation actions.

Example Output:

- Block malicious IP addresses
- Reset compromised credentials
- Enable Multi-Factor Authentication
- Conduct further log analysis

THREAT-1882D9

Severity

High

Risk Score

75

MITRE

T1110

Source IP: 24.151.103.17

Username: elastic_user_0

Attack Type: SSH Brute Force

Analyze THREAT 1882D9

AI Security Analysis

Here is the analysis of the security event:

1. Executive Summary

A high-severity SSH Brute Force attack was detected originating from IP address 24.151.103.17, specifically targeting the `elastic_user_0` account. This incident indicates a concerted effort to gain unauthorized access to a system and poses a significant risk of compromise.

2. Findings

- Attack Type: Confirmed SSH Brute Force attempt (MITRE T1110 - Brute Force).
- Severity & Risk: Rated as High severity with a risk score of 75, indicating a critical threat level.
- Source: The attack originated from external IP 24.151.103.17.
- Target: The `elastic_user_0` account was the specific target of the brute force attempts.
- Status: The event indicates an attempt; it's unclear if the brute force was successful without further log analysis.

3. Recommendations

- Isolate & Block: Immediately block the source IP address (24.151.103.17) at the perimeter firewall.
- Account Management:
 - Temporarily disable the `elastic_user_0` account.
 - If `elastic_user_0` is a valid account, force a password reset and enable Multi-Factor Authentication (MFA). If it's an unnecessary default or generic account, delete it.
- Log Analysis: Review SSH logs for successful logins from the source IP, as well as any other suspicious activity by `elastic_user_0` around the time of the event.
- Harden SSH:
 - Implement stronger password policies.
 - Deploy or configure a brute-force protection tool (e.g., Fail2Ban) to automatically block IPs after multiple failed login attempts.
 - Consider restricting SSH access to known IPs only, or moving SSH to a non-standard port.
- Vulnerability Scan: Conduct an internal vulnerability scan on the targeted system to identify any other potential weaknesses.

9.5 Custom SPL Query Module

Analysts can execute custom Splunk searches directly from the dashboard.

Example Query:

```
search index=* source="ssh_logs_1000.txt"
| stats count
```

Result:

1000 Events

Custom SPL Query

Enter SPL Query

```
search index=* source="ssh_logs_1000.txt"
| stats count
```

Run Query

9.6 Threat Report Export

Threat intelligence data can be exported as CSV reports.

Exported Information:

- Threat ID
- Source IP
- Username
- Severity
- Risk Score
- MITRE Technique
- Attack Type

This feature supports documentation, compliance, and further analysis.

	A	B	C	D	E	F	G	H	I
1	Threat_ID	Source_IP	Username	Severity	Risk_Score	MITRE_ID	Attack_Type	Event_Count	
2	THREAT-1	24.151.10	elastic_user	High	75	T1110	SSH Brute	735	
3	THREAT-2	49.4.143.1	root	Critical	90	T1110	Root Account	300	
4	THREAT-F	34.204.22	invalid	High	75	T1110	SSH Brute	215	
5	THREAT-C	85.245.10	ubuntu	Critical	95	T1078	Successful	175	
6	THREAT-3	181.26.18	invalid	High	75	T1110	SSH Brute	140	
7	THREAT-0	201.178.8	root	Critical	90	T1110	Root Account	120	
8	THREAT-1	122.163.6	invalid	High	75	T1110	SSH Brute	105	
9	THREAT-D	85.245.10	elastic_user	Critical	95	T1078	Successful	105	
10	THREAT-F	85.245.10	elastic_user	Critical	95	T1078	Successful	90	
11	THREAT-5	85.245.10	elastic_user	Critical	95	T1078	Successful	80	
12	THREAT-9	85.245.10	elastic_user	Critical	95	T1078	Successful	80	
13	THREAT-E	85.245.10	elastic_user	Critical	95	T1078	Successful	75	
14	THREAT-2	122.163.6	root	Critical	90	T1110	Root Account	70	
15	THREAT-7	14.54.210	root	Critical	90	T1110	Root Account	70	
16	THREAT-8	181.25.20	invalid	High	75	T1110	SSH Brute	70	
17	THREAT-3	201.178.8	invalid	High	75	T1110	SSH Brute	70	
18	THREAT-F	85.245.10	elastic_user	Critical	95	T1078	Successful	70	
19	THREAT-1	85.245.10	elastic_user	Critical	95	T1078	Successful	55	
20	THREAT-9	85.245.10	elastic_user	Critical	95	T1078	Successful	55	
21	THREAT-2	24.151.10	elastic_user	Critical	95	T1078	Successful	45	
22									

10. Results

The developed SOC dashboard successfully processed and analyzed 1,000 security events.

Key Findings:

- Total Logs Analyzed: 1,000
- Unique Threats Detected: 20
- Critical Threats Identified: 14
- High Threats Identified: 6
- SSH Brute Force Attacks Detected
- Root Account Attacks Detected
- Successful SSH Compromises Identified
- MITRE ATT&CK Mapping Implemented
- AI Investigation Support Successfully Integrated
- CSV Threat Report Export Implemented

The system successfully demonstrated end-to-end SOC functionality.

11. Challenges Faced

During development, several challenges were encountered:

- Splunk API integration issues
- SPL query optimization
- Generative AI quota limitations
- Streamlit session state management

- Threat scoring design and calibration
- MITRE ATT&CK mapping implementation

These challenges were resolved through testing, debugging, and optimization.

12. Future Enhancements

Future improvements may include:

- Real-time log monitoring
- VirusTotal integration
- PDF report generation
- User authentication and role-based access control
- Cloud deployment using AWS

13. Conclusion

The SOC Security Analytics Dashboard successfully demonstrates the integration of SIEM technology, threat intelligence, security analytics, and Generative AI within a unified platform. By leveraging Splunk Enterprise, Python, Streamlit, and Google Gemini AI, the system provides automated threat detection, risk scoring, MITRE ATT&CK mapping, interactive visualizations, and AI-assisted investigations.

The project highlights practical SOC workflows and showcases skills in cybersecurity monitoring, threat analysis, Python development, SIEM operations, and dashboard engineering. The solution provides a scalable foundation for future enhancements such as real-time monitoring, threat intelligence integration, and cloud-based deployment.

Links:

GitHub: <https://github.com/crs-12/AI-Powered-Splunk-Log-Analysis>